

Sex discrimination study:

Takeaways from the playing cards simulation

MATH 213

Sex discrimination study - playing cards simulation

- In the sex discrimination study from class, we saw that in the actual experiment conducted in the 1970's:

Proportion of supervisors
in the male file group
who decided to promote

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Proportion of supervisors
in the female file group
who decided to promote

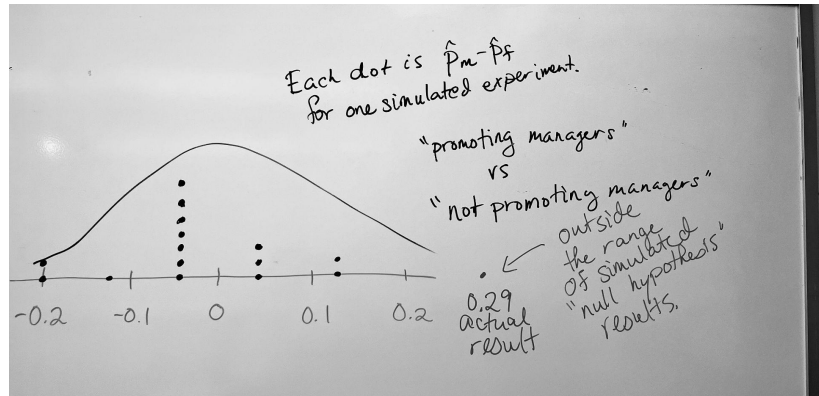
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0.29

- On the face of it, it looks as if discrimination possibly occurred, since 0.29 is a positive number, meaning that a greater proportion of supervisors in the male file group than in the female file group decided to promote.
- A skeptic might wonder: what if no discrimination occurred and in actual fact, just through the random assignment process alone, most of the bank supervisors who might have been inclined to “promote” ended up in the “male-file group” by accident?

- We conducted a simulation of a **randomization hypothesis test**.
- We assumed the “null hypothesis” of no discrimination - that is, the 35 supervisors who decided to promote and the 13 who decided not to promote would have made those same decisions regardless of the gender of the candidate.
- Then we shuffled up our 35 blue cards (representing promoting supervisors) and 13 red cards (representing the not-promoting supervisors) and randomly assigned 24 supervisors to a “male file” group and the remaining 24 to a “female file” group.
- We calculated the difference in proportions of promoting supervisors from each group in the order $p_{\text{male}} - p_{\text{female}}$.
- We did this 18 times in all!
- Here is a dotplot containing the differences of proportions we calculated for our 18 simulations.

Takeaways



- Because the experimental difference of proportions of 0.29 is so far away from the differences we obtained during our simulation, it seems unlikely that the difference of 0.29 occurred simply through the randomization process alone.
- The dotplot gives us a sense of what differences of proportions are likely under the assumption of the null hypothesis of no discrimination. Since our difference of proportions is so far removed from the differences of proportions in the dotplot, it seems unlikely that no discrimination occurred.
- To be more certain of this, we will do another simulation using the computer next time.